# Arduino I/O Demystified

The Arduino interface board has a single row of connectors or pins on each side of the board. These are used mostly for inputs, outputs, and power. Figure 5-1 shows a simple illustration of the Arduino Uno interface board. Your board might be a little different.

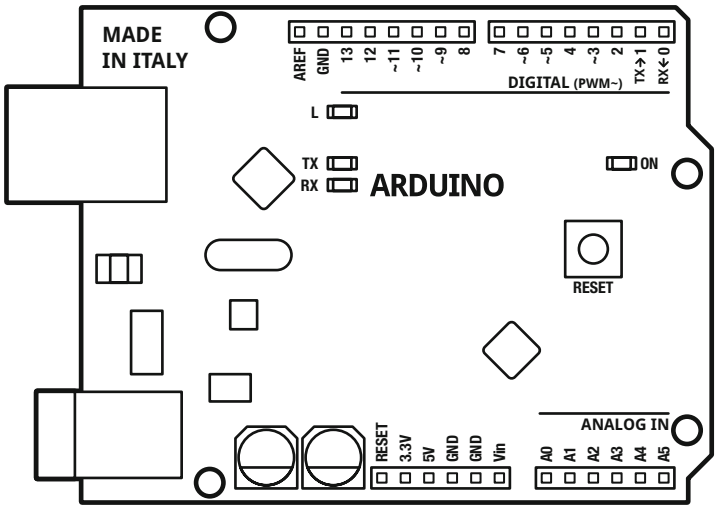



***Figure 5-1.*** *Arduino illustration*

As shown in Figure 5-1 and printed on the interface board, there are 14 pins on one side of the board marked Digital that are numbered from 0 to 13. Of these, two are used for serial communications through USB for programming, debugging, and other forms of communication. These pins are numbered 0 and 1, and are marked RX <- and TX -> respectively. Anything connected to these pins may affect or be affected by serial communications. Pin 13 has a resistor and LED (marked L) connected to it on board so that it can be used for testing, diagnostics, and other forms of blinking. Six of the digital pins can also act as analog outputs using pulse width modulation (PWM). These are pins 3, 5, 6, 9, 10, and 11 on the Arduino Uno and are marked on the interface board with the tilde symbol (~). On the other side of the board are six additional I/O pins labeled as Analog In, numbering A0 through A5. These pins are connected to the microcontroller's analog to digital convertor (ADC) for interpreting incoming analog signals, although they can also be used as additional digital inputs and outputs.

Each I/O pin operates on a range of 0 to +5 VDC, or volts direct current. A range of 0 to +2V is said to be off or `LOW` while anything over about +3V is said to be on or `HIGH`. Generally, a circuit connected to one of the Arduino I/O pins needs to have a high or positive side connected to power (+5 VDC), some form of load, and a low or negative side connected to ground (GND). How a circuit is physically connected to the Arduino or in which direction determines how the I/O pin can be used and which side of the circuit is connected to the Arduino. Each pin can either **source**, or provide a positive biased current, or **sink**, to provide a negative biased current, up to 40 milliamps each. Typically, most Arduino circuits are designed so that they source current, as shown in Figure 5-2, although there are times where sinking current is necessary, as shown in Figure 5-3.